an individual, and then categorize that person as underweight, normal weight, overweight, or obese based on that value.

The body mass index is a value derived from the mass (weight) and height of an individual. The BMI is defined as the body mass divided by the square of the body height and is universally expressed in units of Kg/m^2 resulting from mass in kilograms and height in metres.

$$BMI = \frac{mass_{kg}}{height^2_m}$$

Students Activity

Every student has to calculate their BMI and to be categorized

WHO regards a BMI of less than 18.5 as underweight and may indicate malnutrition. While a BMI equal to or greater than 25% considered overweight and above 30 is considered obese.

CATEGORY	BMI (KG/M2)	
	FROM	TO
Low		18.5
Normal	18.5	25
Obese (level 1)	25	30
Obese (level 2)	30	35
Obese (level 3)	35	40
Obese (level 4)	40	

BMI ranges are based on the relationship between body weight and disease and death.

Overweight and obese individuals are at an increased risk for the following diseases

- Coronary artery disease
- Dyslipidemia
- Type 2 diabetes
- Gall bladder disease
- Hypertension
- Osteoarthritis
- Stroke

5.5 Procedure and Recording of Temperature

Body temperature is its degree of heat. Normally the degree at which balance between heat production and heat loss is maintained is called the "Normal Body Temperature". It is also called as "Normothermia or Euthermia".



Adult Normal Temperature is 37°C (98.4°F)

5.5.1 Factors affecting body temperature

1) Age, 2) Exercise, 3) Hormone level, 4) circadian rhythm 5) Stress, 6) Environment, 7) Temperature attraction, 8) Fever, 9) Hyperpyrexia, 10) Heat stroke, 11) Hypothermia

1) **Age:** Temperature regulation is unstable until children reach puberty. Older adults are sensitive to temperature extremes because of deterioration in control mechanisms, reduced sweat gland activity, reduced amounts of subcutaneous fat and reduced metabolism

2) **Exercise:** Muscle activity causes increased metabolism by increasing carbohydrate and fat breakdown.

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